

GOSFORD HIGH SCHOOL

2018 HIGHER SCHOOL CERTIFICATE TRIAL EXAMINATION

Mathematics Extension 2

- **General Instructions**
- Reading time – 5 minutes
- Working time – 3 hours
- Write using black pen
- Board-approved calculators may be used
- A reference sheet is provided at the back of this paper
- In Questions 11 – 16, show relevant mathematical reasoning and/or calculations

Total Marks – 100

Section I Pages 2 – 6

10 marks

- Attempt Questions 1 – 10
- Allow about 15 minutes for this section

Section II Pages 7 – 14

90 marks

- Attempt Questions 11 – 16
- Allow about 2 hours and 45 minutes for this section

Section I

10 marks

Attempt Questions 1 – 10.

Allow about 15 minutes for this section.

Use the multiple-choice answer sheet for Questions 1 – 10.

1. An ellipse has Cartesian equation $\frac{x^2}{4} + \frac{y^2}{2} = 1$.

What is the parametric equation of the ellipse?

- (A) $x = 2\cos\theta$, $y = \sqrt{2}\sin\theta$
- (B) $x = 4\cos\theta$, $y = 2\sin\theta$
- (C) $x = \sqrt{2}\sin\theta$, $y = 2\cos\theta$
- (D) $x = 2\sin\theta$, $y = 4\cos\theta$
2. What is the square root of $12 - 16i$?
- (A) $\pm(2 - 4i)$
- (B) $\pm(2\sqrt{3} - 4i)$
- (C) $\pm(4 - 2i)$
- (D) $\pm(4 - 2\sqrt{3}i)$

3. The region bounded by the curve $y = x^2$, the x -axis, $x = 0$ and $x = 2$ is rotated around the line $x = 2$.

Which of the following gives the volume of the solid formed?

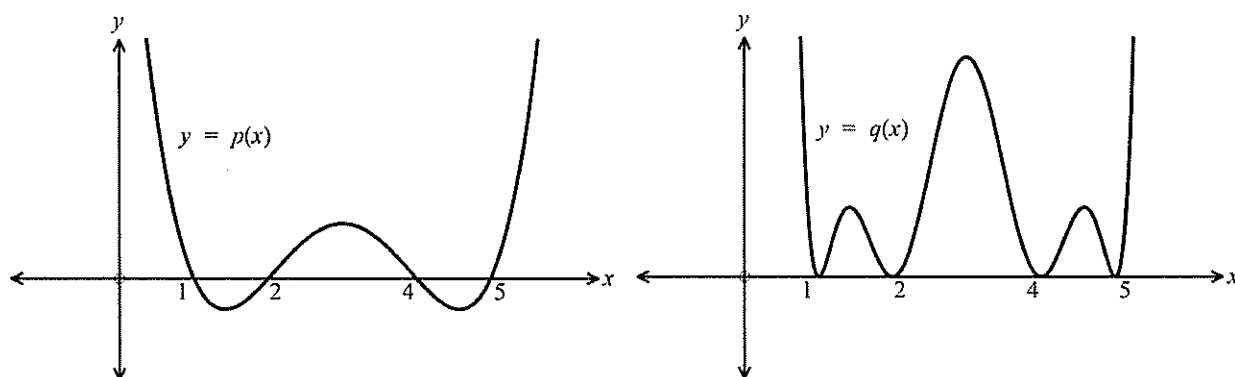
(A) $V = \pi \int_0^2 (2-x)x^2 dx$

(B) $V = \pi \int_0^4 (2-x)x^2 dx$

(C) $V = 2\pi \int_0^2 (2-x)x^2 dx$

(D) $V = 2\pi \int_0^2 x^2 (2-x)^2 dx$

4. The graphs of two functions, $y = p(x)$ and $y = q(x)$ are drawn below.



Which of the following describes the relationship between the two functions?

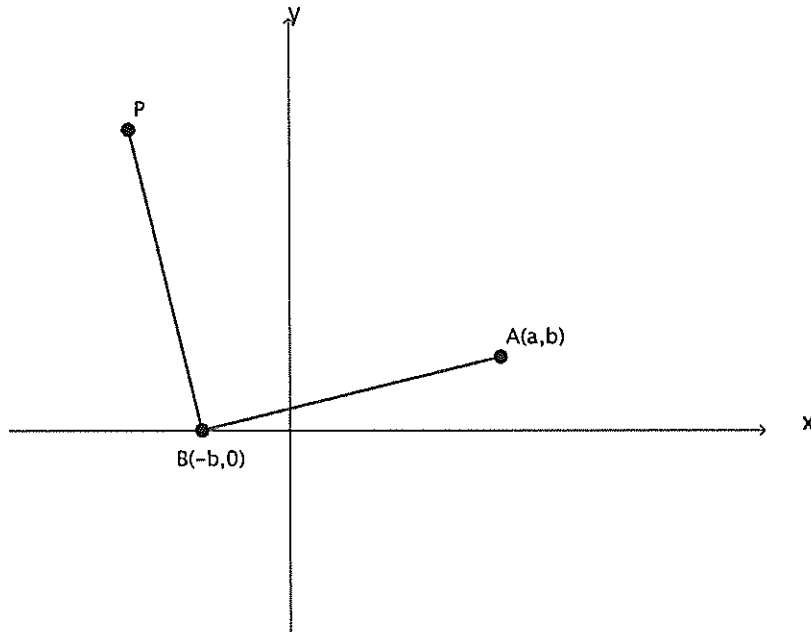
(A) $q(x) = \frac{1}{p(x)}$

(B) $q(x) = [p(x)]^2$

(C) $p(x) = \frac{1}{q(x)}$

(D) $p(x) = [q(x)]^2$

5.



The Argand diagram above shows the point $A(a, b)$ representing the complex number $z = a + ib$, where a and b are real. B is the point $(-b, 0)$.

P is a point such that $PB = 2 \times AB$ and $\angle ABP = 90^\circ$.

Which of the following complex numbers does P represent?

- (A) $-2b + i(2a)$
- (B) $-b + ai$
- (C) $-2b + i(2a + 2b)$
- (D) $-3b + i(2a + 2b)$

6. Given $\lim_{x \rightarrow \infty} \left(1 + \frac{1}{x}\right)^x = e$, which of the following is the value of $\lim_{x \rightarrow \infty} \left(\frac{x+4}{x-1}\right)^{x+4}$?

- (A) e^5
- (B) $e^5 + 1$
- (C) e^6
- (D) $e^6 + 1$

7. What is the value of the constant B such that $P(x) = (x - \alpha)^2 Q(x) + Ax + B$?
- (A) $B = P(\alpha)$
- (B) $B = P'(\alpha)$
- (C) $B = P(\alpha) - \alpha P'(\alpha)$
- (D) $B = P'(\alpha) - \alpha P(\alpha)$
8. Solve the inequality $\frac{x+1}{x-3} \leq \frac{x+3}{x-2}$
- (A) $x < 2$ and $x > 3$
- (B) $x < 2$ and $3 < x \leq 7$
- (C) $2 < x < 3$
- (D) $2 < x < 3$ and $x \geq 7$
9. What is the value of the constant k such that the function, $f(x)$, is continuous at $x=0$, where $f(x)$ is defined by:
- $$f(x) = \frac{\sqrt{x+1} - 1}{x} \text{ for } x \neq 0$$
- and $f(0) = k$, at $x=0$
- (A) $k = -1$
- (B) $k = 0$
- (C) $k = \frac{1}{2}$
- (D) $k = 1$

10. $P\left(cp, \frac{c}{p}\right)$ and $Q\left(cq, \frac{c}{q}\right)$, where $p \neq 0$ and $q \neq 0$, are two points on the rectangular hyperbola $xy = c^2$.

What is the condition for the tangent to the hyperbola at P to be perpendicular to the line OQ ?

(A) $|pq| = 1$

(B) $p^2q = 1$

(C) $pq^2 = 1$

(D) $p^2 - q^2 = 1$

Section II

90 marks

Attempt Questions 11 – 16.

Allow about 2 hours and 45 minutes for this section.

Answer each question in a **SEPARATE** writing booklet. Extra writing booklets are available.

In Questions 11 – 16, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks) Use the Question 11 writing booklet.

(a) If $z = 2 + i$ and $w = 4 - i$ find in the form $a + ib$, where a and b are real, the values of:

(i) $\overline{2z - w}$ 1

(ii) $\frac{w}{z}$ 2

(b) Find $\int \frac{e^{3x} + 8}{e^x + 2} dx$ 2

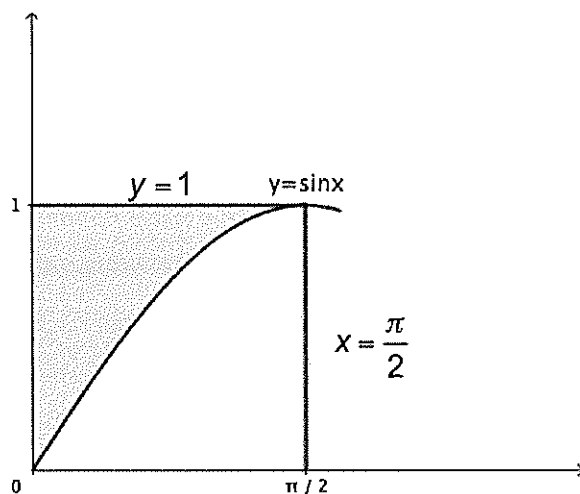
(c) The equation $x^3 - 5x^2 + 3x - 2 = 0$ has roots α, β and γ . 3
Find a cubic equation with integer coefficients that has roots α^2, β^2 and γ^2 .

(d) Use the substitution $u = \cos 2x$ to find $\int \cos^2 2x \sin^3 2x dx$. 3

Question 11 continues on page 8

Question 11 continued

(e)



In the diagram, the area above the curve $y = \sin x$, between the lines $x = 0$ and $x = \frac{\pi}{2}$, is rotated about the line $y = 1$.

(i) Use discs formed by slicing perpendicular to the line $y = 1$ to show that the solid

1

formed is given by $V = \pi \int_0^{\frac{\pi}{2}} (1 - 2 \sin x + \sin^2 x) dx$.

(ii) Find the value of V in simplest exact form.

3

End of Question 11

Question 12 (15 marks) Use the Question 12 writing booklet.

(a) z is a complex number such that $\left|z - 2\sqrt{2}(1+i)\right| = 2$.

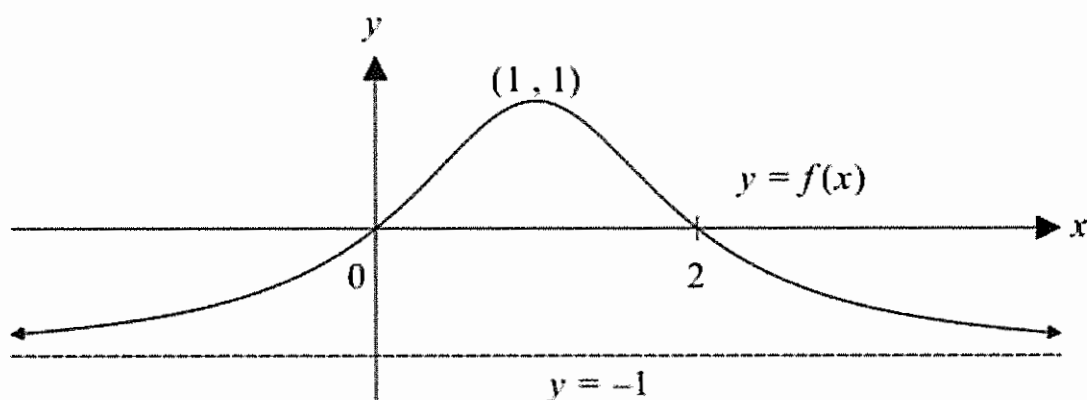
(i) Sketch the locus of the point P representing the complex number z in an Argand diagram. 2

(ii) Q is the point on the locus where z has its smallest principal argument. Find the value of the complex number represented by Q in mod-arg form. 2

(b) The hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ (where a and b are positive) has a focus at the point $(3\sqrt{2}, 0)$ and the line $y = \frac{2x}{\sqrt{5}}$ is an asymptote. 3

Find the values of a and b .

(c)



The diagram shows the graph of the function $f(x) = \frac{x(2-x)}{x^2-2x+2}$.

On separate diagrams sketch the graphs of the following curves, clearly showing the intercepts on the axes and the equations of any asymptotes.

(i) $y = -f(|x|)$ 2

Question 12 continues on page 10

Question 12 continued

(ii) $y = \frac{1}{f(x)}$ **2**

(iii) $y = \log_e f(x)$ **2**

(iv) $y = f'(x)$ **2**

End of Question 12

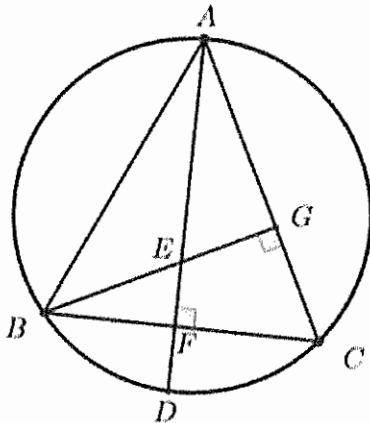
Question 13 (15 marks) Use the Question 13 writing booklet.

- (a) The equation of a curve is $x^3 - 3x^2y + y^3 = 3$.
- (i) Show that $\frac{dy}{dx} = \frac{x^2 - 2xy}{x^2 - y^2}$ 2
- (ii) Find the coordinates of the points on the curve where the tangent is parallel to the x -axis. 3
- (b) Let α, β, γ be the non-zero roots of the equation $x^3 + rx + s = 0$.
- (i) Find, in terms of s , the simplified value of $\alpha^3 + \beta^3 + \gamma^3$ 2
- (ii) If $x^3 + rx + s = 0$ has a double root, show that $x = -\frac{3s}{2r}$ 2
- (c) A solid has an elliptical base with equation $4x^2 + 25y^2 = 100$. Each cross section perpendicular to the x -axis is a right angled isosceles triangle with the hypotenuse in the base of the solid.
- (i) Draw a diagram representing the elliptical base, showing all intercepts with the axes. 2
- (ii) Find the volume of the solid. 4

End of Question 13

Question 14 (15 marks) Use the Question 14 writing booklet.

(a)



The diagram above shows triangle ABC inscribed in a circle.

G is the point on AC such that BG is perpendicular to AC and F is the point on BC such that AF is perpendicular to BC .

AF and BG meet at E .

AF produced meets the circle at D .

(i) Explain why $ABFG$ is a cyclic quadrilateral. 1

(ii) Prove that $DF = EF$. 3

(b) (i) Show that $\frac{1}{(2t+1)(t+2)} = \frac{2}{3(2t+1)} - \frac{1}{3(t+2)}$ 2

(ii) Use the substitution $t = \tan \frac{x}{2}$ to evaluate in simplest exact form $\int_0^{\frac{\pi}{2}} \frac{dx}{4+5 \sin x}$. 3

Question 14 continues on page 13

Question 14 continued

- (c) $P\left(3p, \frac{3}{p}\right)$ and $Q\left(3q, \frac{3}{q}\right)$ are points on the rectangular hyperbola $xy = 9$.
- (i) Show that the equation of the chord PQ is $x + p q y = 3(p + q)$ 2
- (ii) Find the co-ordinates of N , the midpoint of PQ . 1
- (iii) If the chord PQ is a tangent to the parabola $y^2 = 3x$, prove that the locus of N is $3x = -8y^2$. 3

End of Question 14

Question 15 (15 marks) Use the Question 15 writing booklet.

- (a) Initially a speedboat is travelling at a speed of 15 ms^{-1} in a straight line across a lake. At time t seconds later, the speedboat has a velocity $v \text{ ms}^{-1}$ and the engine is producing a constant force of 600 Newtons.

The speedboat experiences a resistance force of magnitude $90v$ Newtons.

The mass of the speedboat plus passengers is 450 kg.

Assume the water in the lake is still.

(i) Show that $\frac{dv}{dt} = -\frac{3v-20}{15}$ 1

(ii) Find an expression for v in terms of t . 3

(iii) Find the time taken for the speed of the speedboat to reduce to 10 ms^{-1} . 2

(b) Given that $I_n = \int_0^3 x^n \sqrt{9-x^2} dx$, $n = 0, 1, 2, \dots$

(i) Show that $(n+2)I_n = 9(n-1)I_{n-2}$, $n = 2, 3, 4, \dots$ 3

(ii) Find the value of I_4 2

(c) Given that $w+x+y+z = \pi$:

(i) Show that $\sin z = \sin(w+x)\cos y + \cos(w+x)\sin y$. 1

(ii) Hence show that $\sin w \sin y + \sin x \sin z = \sin(w+x)\sin(x+y)$. 3

End of Question 15

Question 16 (15 marks) Use the Question 16 writing booklet.

- (a) (i) Use de Moivre's theorem to show that $\frac{\sin 7\theta}{\sin \theta} = 7 - 56 \sin^2 \theta + 112 \sin^4 \theta - 64 \sin^6 \theta$. 3
- (ii) Explain why $\sin^2 \frac{\pi}{7}$ is a root of the equation $64x^3 - 112x^2 + 56x - 7 = 0$ and write down the two other roots in trigonometric form. 2
- (iii) Hence show that the value of $\operatorname{cosec}^2 \frac{\pi}{7} + \operatorname{cosec}^2 \frac{2\pi}{7} + \operatorname{cosec}^2 \frac{3\pi}{7} =$ 1
- (b) (i) Using the binomial theorem, write down the expansion of $(1+i)^{2m}$, where $i = \sqrt{-1}$ and m is a positive integer. 1
- (ii) Hence show that 2
- $$1 - \binom{2m}{2} + \binom{2m}{4} - \binom{2m}{6} + \dots + (-1)^m \binom{2m}{2m} = 2^m \cos\left(\frac{1}{2}\pi m\right),$$
- where m is a positive integer.
- (c) A particle of mass m kg is projected vertically upwards with speed U ms^{-1} .
 At time t seconds the particle has vertical height x metres above the point of projection, speed v ms^{-1} and acceleration a ms^{-2} .
 The particle moves vertically under gravity in a medium where the resistance to motion has magnitude $\frac{mv^2}{g}$ Newtons, where g ms^{-2} is the acceleration due to gravity.
- (i) Show that $a = -\frac{1}{g}(g^2 + v^2)$. 1
- (ii) Show that $v = g\left(\frac{U - g \tan t}{g + U \tan t}\right)$ and find the time taken for the particle to reach its greatest height. 3
- (iii) Express x in terms of t . 2

End of Paper

EXT 2 TRIAL 2018

m-c

1) $\frac{x^2}{4} + \frac{y^2}{2} = 1$

$x = 2 \cos \theta$ (A)

$y = \sqrt{2} \sin \theta$

(ie $\frac{4 \cos^2 \theta}{4} + \frac{2 \sin^2 \theta}{2} = 1$)

2) $12 - 16i$

$(x + iy)^2 = 12 - 16i$

$x^2 - y^2 = 12$

$2xy = -16 \rightarrow y = \frac{-8}{x}$

$x^2 - \frac{64}{x^2} = 12$

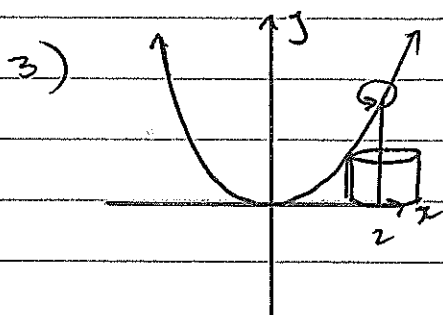
$x^4 - 12x^2 - 64 = 0$

$(x^2 - 16)(x^2 + 4) = 0$

$x = \pm 4$

$x = 4 \quad y = -2 \quad 4 - 2i$ (C)

$x = -4 \quad y = 2 \quad -4 + 2i$



$V = 2\pi \int_0^2 (2 - x^2) x^2 dx$ (C)

4) (B)

5) $\vec{BA} = \vec{BO} + \vec{OA}$

$= b + a + bi$

$\vec{BP} = (\vec{BA}) 2i$

$= 2i(a + b) - 2b$

$\vec{OP} = \vec{OB} + \vec{BP}$

$= -b - 2b + 2i(a + b)$

$= -3b + 2i(a + b)$ (D)

6) $\lim_{x \rightarrow \infty} \left(1 + \frac{1}{x}\right)^x = e$

Find $\lim_{x \rightarrow \infty} \left(\frac{x+4}{x-1}\right)^{x+4}$

Let $\frac{x+4}{x-1} = 1 + \frac{1}{y}$

$\therefore \frac{1}{y} = \frac{x+4 - x + 1}{x-1}$

$\frac{1}{y} = \frac{5}{x-1}$

$\therefore x = 5y + 1$

$\lim_{x \rightarrow \infty} \left(\frac{x+4}{x-1}\right)^{x+4}$

$= \lim_{y \rightarrow \infty} \left(\frac{5y+5}{5y}\right)^{5y+5}$

$= \lim_{y \rightarrow \infty} \left(1 + \frac{1}{y}\right)^{5y+5}$

$= \lim_{y \rightarrow \infty} \left(\left(1 + \frac{1}{y}\right)^y\right)^5 \left(1 + \frac{1}{y}\right)^5$

$= e^5 (1)$

$= e^5$ (A)

Q11

$$2) \quad z = 2+i \quad w = 4-i$$

$$i) \quad 2z - w = 4 + 2i - 4 + i \\ = 3i$$

$$\overline{2z - w} = -3i \quad (1)$$

$$ii) \quad \frac{w}{z} = \frac{4-i}{2+i} \times \frac{2-i}{2-i} \quad \perp \\ = \frac{8 - 6i - 1}{4 + 1}$$

$$= \frac{7}{5} - \frac{6}{5}i \quad \perp \quad (2)$$

$$b) \quad \int \frac{e^{3x} + 8}{e^x + 2} dx$$

$$= \int \frac{(e^x)^3 + 2^3}{e^x + 2}$$

$$= \int \frac{(e^x + 2)(e^{2x} - 2e^x + 4)}{e^x + 2} dx$$

$$= \int e^{2x} - 2e^x + 4 dx$$

$$= \frac{1}{2} e^{2x} - 2e^x + 4x + C \quad \perp$$

(2)

$$c) \quad x^3 - 5x^2 + 3x - 2 = 0$$

$$x = 2^2 \quad 2 = \sqrt{x}$$

$$(\sqrt{x})^3 - 5(\sqrt{x})^2 + 3\sqrt{x} - 2 = 0$$

$$x\sqrt{x} + 3\sqrt{x} = 5x + 2$$

$$(x\sqrt{x} + 3\sqrt{x})^2 = (5x + 2)^2 \quad \perp$$

$$x^3 + 6x^2 + 9x = 25x + 20x + 4$$

$$x^3 - 19x^2 - 11x - 4 = 0 \quad \perp$$

(3)

$$d) \quad \int \cos^2 2x \sin^3 2x dx$$

$$u = \cos 2x$$

$$\frac{du}{dx} = -2 \sin 2x$$

$$dx = \frac{du}{-2 \sin 2x}$$

$$I = \int u^2 \cdot \frac{1}{-2} (1 - u^2) du$$

$$= -\frac{1}{2} \int u^2 - u^4 du$$

$$= -\frac{1}{2} \left(\frac{u^3}{3} - \frac{u^5}{5} \right) + C$$

$$= \frac{\cos^3 2x}{10} - \frac{\cos^5 2x}{6} + C \quad (3)$$

$$e) \quad i) \quad \Delta V = \pi (1-y)^2 \delta x \quad (1)$$

$$V = \int_0^{\pi/2} \pi (1 - \sin x)^2 dx$$

$$= \pi \int_0^{\pi/2} (1 - 2\sin x + \sin^2 x) dx$$

$$ii) \quad V = \pi \int_0^{\pi/2} (1 - 2\sin x + (\frac{1}{2} - \frac{1}{2} \cos 2x)) dx$$

$$= \pi \int_0^{\pi/2} \left(\frac{3}{2} - 2\sin x - \frac{1}{2} \cos 2x \right) dx$$

$$= \pi \left[\frac{3x}{2} + 2\cos x - \frac{\sin 2x}{4} \right]_0^{\pi/2}$$

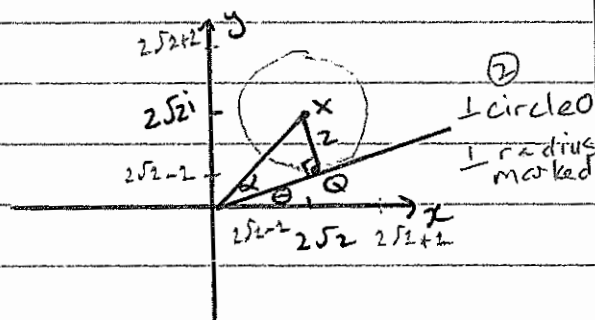
$$= \pi \left(\left(\frac{3\pi}{4} + 0 \right) - (2) \right)$$

$$= \frac{3\pi^2}{4} - 2\pi \quad \perp \quad (3)$$

Q12

i) $|3 - (2\sqrt{2} + 2\sqrt{2}i)| = 2$

i)



ii) $OX = \sqrt{8+8} = 4$

$\sin \alpha = \frac{1}{2} \therefore \alpha = \frac{\pi}{6}$

$4 + \tan(\alpha + \theta) = 1$

$\therefore \alpha + \theta = \frac{\pi}{4}$

$\therefore \theta = \frac{\pi}{4} - \frac{\pi}{6}$

$\theta = \frac{\pi}{12}$

$OQ = \sqrt{16-4}$

$= \sqrt{12} = 2\sqrt{3}$

$\therefore Q: 2\sqrt{3} \text{ cis } \frac{\pi}{12}$ ②

c) $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$

$S(3\sqrt{2}, 0)$ asympt $y = \frac{2x}{\sqrt{5}}$

$\therefore \frac{b}{a} = \frac{2}{\sqrt{5}} \quad ae = 3\sqrt{2}$

$b = \frac{2a}{\sqrt{5}} \quad e = \frac{3\sqrt{2}}{a}$

$b^2 = a^2(e^2 - 1)$

$\frac{4a^2}{5} = a^2 \left(\frac{18}{a^2} - 1 \right)$

$\frac{4a^2}{5} = 18 - a^2$

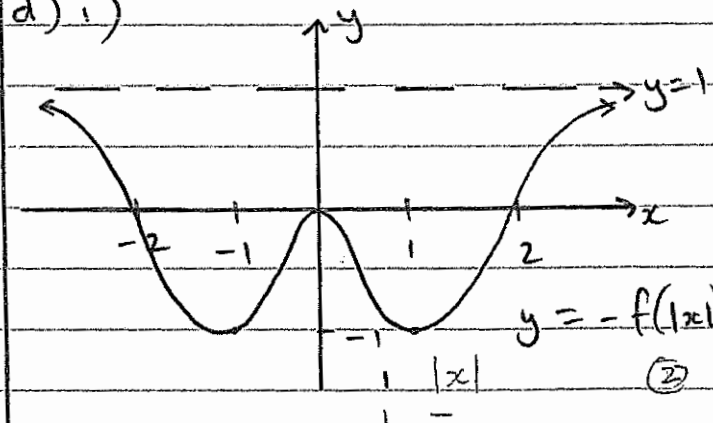
$9a^2 = 90$

$a^2 = 10$

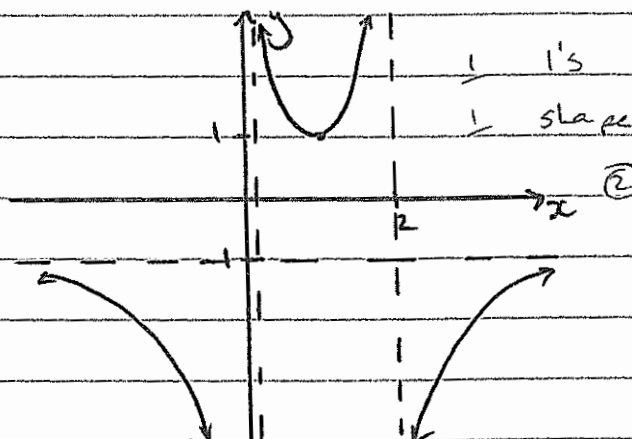
$a = \sqrt{10}$

$b = 2\sqrt{2}$ ③

d) i)



ii)



Q13

$$a) x^3 - 3x^2y + y^3 = 3$$

$$i) 3x^2 - y \cdot 6x - 3x^2 \frac{dy}{dx} + 3y^2 \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} (3x^2 - 3y^2) = 3x(x - 2y)$$

$$\frac{dy}{dx} = \frac{3x(x - 2y)}{3(x^2 - y^2)}$$

$$= \frac{x^2 - 2xy}{x^2 - y^2}$$

! has same

! answer

(2)

ii) parallel to x-axis

$$\frac{dy}{dx} = 0$$

$$\therefore x(x - 2y) = 0$$

$$\left(\begin{array}{l} x=0 \\ y=\sqrt[3]{3} \end{array} \right) \quad x=2y \quad !$$

$$8y^3 - 12y^3 + y^3 = 3$$

$$-3y^3 = 3$$

$$y^3 = -1$$

$$y = -1$$

$$x = -2$$

 $\therefore$ pts parallel to x-axis

$$(0, \sqrt[3]{3}), (-2, -1) \quad (3)$$

$$b) x^3 + rx + s = 0$$

$$i) x^3 = -rx - s \quad \text{let } x = \alpha$$

$$\alpha^3 = -r\alpha - s \quad \beta, \gamma$$

$$\beta^3 = -r\beta - s \quad !$$

$$\gamma^3 = -r\gamma - s$$

$$\underline{\underline{\alpha^3 = -r(2\alpha) - 3s}}$$

$$\underline{\underline{2\alpha = 0}}$$

(2)

$$\therefore \alpha^3 + \beta^3 + \gamma^3 = -3s \quad !$$

$$ii) P(x) = x^3 + rx + s$$

$$P'(x) = 3x^2 + r$$

if double root

$$P(x) = P'(x) = 0$$

$$\therefore x^3 + rx + s = 3x^2 + r = 0$$

$$\text{let } x^2 = -\frac{r}{3}$$

$$\therefore x(-\frac{r}{3}) + rx + s = 0$$

$$x(r - \frac{r}{3}) = -s$$

$$\frac{d}{dx} \frac{dy}{dx}$$

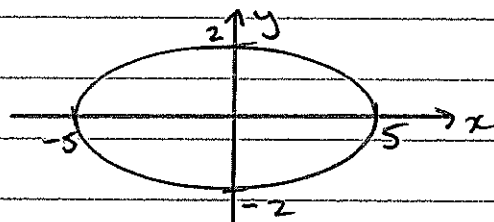
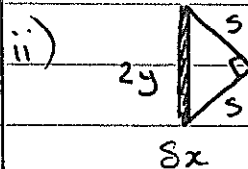
$$x(\frac{2r}{3}) = -s$$

$$x = -\frac{3s}{2r} \quad ! \quad (2)$$

$$c) 4x^2 + 25y^2 = 100$$

$$i) \frac{x^2}{25} + \frac{y^2}{4} = 1$$

(2)

! intercepts
! shape

8x

$$2s^2 = 4y^2$$

$$s^2 = 2y^2$$

$$A = \frac{1}{2} s^2$$

$$= \frac{1}{2} \cdot 2y^2$$

$$= y^2 \quad !$$

$$y^2 = 4 - \frac{4}{25} x^2$$

$$V = 2 \int_0^5 4 - \frac{4}{25} x^2 dx \quad !$$

$$= 2 \left[4x - \frac{4x^3}{75} \right]_0^5$$

$$= 8 \left[(5 - \frac{125}{75}) - 0 \right] \quad (4)$$

$$= \frac{80}{3} \quad !$$

Q14

a) i) $\angle AFB = \angle AGB = 90$
 (given $BG \perp AC$, $AF \perp BC$)
 $\therefore ABFG$ is cyclic quad
 (= $\angle$'s at circumference
 from same arc AB) (1)

ii) $\angle BFE = \angle BFD = 90$
 ($AF \perp BC$, given)
 $\angle FBG = \angle FAG$
 (= $\angle$'s at circumference
 from same arc FG)
 $\angle FAG = \angle DAC$
 $\angle DAC = \angle DBC$
 (= $\angle$'s at circumference
 from same arc DC)
 $\therefore \angle DBC = \angle DBF$
 $\angle DBF = \angle FBE$

$\therefore$
 In $\triangle BFE + \triangle BFD$
 BF is common side
 $\therefore \triangle BFE \cong \triangle BFD$ (SAA)
 $\therefore DF = EF$
 (corresponding sides
 in congruent $\triangle$'s) (3)

b) i) $\frac{1}{(2t+1)(t+2)} = \frac{a}{2t+1} + \frac{b}{t+2}$

$\therefore a(t+2) + b(2t+1) = 1$
 equate coeffs.

$a + 2b = 0 \quad a = -2b$
 $2a + b = 1$

$\therefore -4b + b = 1$

$-3b = 1$

$b = -\frac{1}{3}$

$a = \frac{2}{3}$

$\therefore \frac{1}{(2t+1)(t+2)} = \frac{2}{3(2t+1)} - \frac{1}{3(t+2)}$ (2)

ii) $\int_0^{\pi/2} \frac{dx}{4+5\sin x}$

$t = \tan \frac{x}{2}$

$x = 2 \tan^{-1} t$

$\frac{dx}{dt} = \frac{2}{1+t^2}$

$x = \frac{\pi}{2} \quad t = 1$

$x = 0 \quad t = 0$

$I = \int_0^1 \frac{2}{1+t^2} \frac{dt}{4+5(\frac{2t}{1+t^2})}$

$= \int_0^1 \frac{2 dt}{4+4t^2+10t}$

$= \int_0^1 \frac{dt}{2t^2+5t+2}$

$= \int_0^1 \frac{dt}{(2t+1)(t+2)}$

$= \int_0^1 \frac{2}{3(2t+1)} - \frac{1}{3(t+2)} dt$

$= \frac{1}{3} [\ln(2t+1) - \ln(t+2)]_0^1$

$= \frac{1}{3} \left[\ln \frac{2t+1}{t+2} \right]_0^1$

$= \frac{1}{3} \left[\ln \frac{3}{2} - \ln \frac{1}{2} \right]$

$= \frac{1}{3} \ln 2$ (3)

Q15

a) at $t=0$ $v=15$ i) $\square \rightarrow 600\text{N}$ $\overleftarrow{90\text{V}}$ $m=450$

$$450a = 600 - 90\text{V}$$

$$a = \frac{20 - 3\text{V}}{15}$$

$$\frac{dv}{dt} = -\frac{3\text{V} - 20}{15} \quad (1)$$

$$\text{ii) } \frac{dt}{dv} = \frac{-15}{3\text{V} - 20}$$

$$t = -5 \ln(3\text{V} - 20) + C$$

$$\text{at } t=0 \quad v=15$$

$$\therefore 0 = -5 \ln 25 + C$$

$$C = 5 \ln 25$$

$$\therefore t = -5 \ln(3\text{V} - 20) + 5 \ln 25$$

$$t = 5 \ln \frac{25}{3\text{V} - 20}$$

$$e^{\frac{t}{5}} = \frac{25}{3\text{V} - 20}$$

$$3\text{V} - 20 = \frac{25}{e^{t/5}}$$

$$v = \frac{25 + 20e^{t/5}}{3e^{t/5}} \quad (3)$$

iii) find t at $v=10$

$$10 = \frac{25 + 20e^{t/5}}{3e^{t/5}}$$

$$30e^{t/5} - 20e^{t/5} = 25$$

$$10e^{t/5} = 25$$

$$e^{t/5} = 5/2$$

$$\frac{t}{5} = \ln\left(\frac{5}{2}\right)$$

$$t = 5 \ln\left(\frac{5}{2}\right)$$

$$t \approx 4.6\text{s} \quad (2)$$

$$\text{b) } I_n = \int_0^3 x^n \sqrt{9-x^2} dx$$

$$\text{i) } = \int_0^3 x^{n-1} \cdot x \sqrt{9-x^2} dx$$

$$u = x^{n-1}$$

$$v' = x \sqrt{9-x^2}$$

$$u' = (n-1)x^{n-2}$$

$$v = -\frac{1}{2}(9-x^2)^{3/2}$$

$$v = -\frac{1}{3}(9-x^2)^{3/2}$$

$$I_n = -\frac{1}{3} \left[x^{n-1} (9-x^2)^{3/2} \right]_0^3$$

$$+ \frac{1}{3} \int_0^3 (9-x^2)^{3/2} (n-1)x^{n-2} dx$$

$$= \frac{n-1}{3} \int_0^3 x^{n-2} \sqrt{9-x^2} (9-x^2) dx$$

$$= \frac{n-1}{3} \int_0^3 9x^{n-2} \sqrt{9-x^2} - x^n \sqrt{9-x^2} dx$$

$$= \frac{n-1}{3} (9I_{n-2} - I_n)$$

$$I_n \left(1 + \frac{n-1}{3}\right) = 3(n-1)I_{n-2}$$

$$I_n \left(\frac{3+n-1}{3}\right) = 3(n-1)I_{n-2}$$

$$(n+2)I_n = 9(n-1)I_{n-2}$$

$$\text{ii) } I_4 = \frac{9}{6}(3)I_2$$

$$= \frac{9}{2} \left(\frac{9}{4} I_0 \right)$$

$$= \frac{81}{8} I_0 \quad (3)$$

Q16

a) i) $(\cos \theta)^7 = \cos 7\theta$

+ $(\sin \theta)^7$

$$= \binom{7}{0} c^7 + \binom{7}{1} i s c^6 - \binom{7}{2} s^2 c^5$$

$$- \binom{7}{3} i s^3 c^4 + \binom{7}{4} s^4 c^3$$

$$+ \binom{7}{5} i s^5 c^2 - \binom{7}{6} s^6 c$$

$$- \binom{7}{7} i s^7$$

equate Im parts

$\therefore \sin 7\theta$

$$= 7 \sin \theta \cos^6 \theta - 35 \sin^3 \theta \cos^4 \theta$$

$$+ 21 \sin^5 \theta \cos^2 \theta - \sin^7 \theta$$

$(\div \sin \theta)$

$$\frac{\sin 7\theta}{\sin \theta} = 7 \cos^6 \theta - 35 \sin^2 \theta \cos^4 \theta$$

$$+ 21 \sin^4 \theta \cos^2 \theta - \sin^6 \theta$$

$$= 7(1 - \sin^2 \theta)^3 - 35 \sin^2 \theta (1 - \sin^2 \theta)^2$$

$$+ 21 \sin^4 \theta (1 - \sin^2 \theta) - \sin^6 \theta$$

$$= 7(1 - 3\sin^2 \theta + 3\sin^4 \theta - \sin^6 \theta)$$

$$- 35s^2(1 - 2s^2 + s^4)$$

$$+ 21s^4 - 21s^6 - s^6$$

$$= 7 - 21s^2 + 21s^4 - 7s^6$$

$$- 35s^2 + 70s^4 - 35s^6$$

$$+ 21s^4 - 21s^6 - s^6$$

$$= 7 - 56\sin^2 \theta + 112\sin^4 \theta$$

$$- 64\sin^6 \theta \quad \text{--- (3)}$$

ii) let $x = \sin^2 \frac{\pi}{7}$

$$\therefore 0 = 7 - 56x + 112x^2 - 64x^3$$

$$\text{ie } 64x^3 - 112x^2 + 56x - 7 = 0$$

N.B $\sin \theta \neq 0$

+ $\sin 7\theta = 0$

$$\therefore 7\theta = \pi k \quad \text{--- (4)}$$

$$\theta = \frac{\pi k}{7} \quad k=1,2,3$$

$$\therefore \text{roots are } \sin^2 \frac{\pi}{7}, \sin^2 \frac{2\pi}{7},$$

$$\sin^2 \frac{3\pi}{7} \quad \text{--- (2)}$$

$$\text{iii) } \operatorname{cosec}^2 \frac{\pi}{7} + \operatorname{cosec}^2 \frac{2\pi}{7} + \operatorname{cosec}^2$$

$$= \frac{1}{\sin^2} + \frac{1}{\sin^2} + \frac{1}{\sin^2}$$

$$= \frac{\Sigma \times \Sigma}{2 \times \Sigma}$$

$$= \frac{56}{64}$$

$$\frac{7}{64}$$

$$= 8 \quad \text{--- (1)}$$

b) i) $(1+i)^{2m}$

$$= (\sqrt{2} \operatorname{cis} \frac{\pi}{4})^{2m}$$

By Binomial theorem

$$= \binom{2m}{0} + \binom{2m}{1} i - \binom{2m}{2} - \binom{2m}{3} i$$

$$\dots + \binom{2m}{2m} (-1)^m \quad \text{--- (1)}$$

ii) Equate real parts

$$1 - \binom{2m}{2} + \binom{2m}{4} - \dots + \binom{2m}{2m} (-1)^m$$

$$= 2^m \cos \frac{\pi m}{2} \quad \text{--- (2)}$$

$$(\sqrt{2} \operatorname{cis} \frac{\pi}{4})^{2m}$$